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library(tidyverse)
library(lme4)
library(lmerTest)
library(data.table)

print("1. Loading RT Data...")

## [1] "1. Loading RT Data..."

rt_data <- fread("/Users/chenyanze/NS-analysis/naturalstories/naturalstories_RTS/processed_RTs.tsv")

print("2. Loading Freq data and performing Logarithmic calculation")

## [1] "2. Loading Freq data and performing Logarithmic calculation"

freq1_data <- fread("/Users/chenyanze/NS-analysis/naturalstories/freqs/freqs-1.tsv", header = FALSE,
                    col.names = c("token_code", "n_order", "word", "freq_n", "freq_n_minus_1")) %>%
  filter(grepl("\\.word$", token_code)) %>% # Only keep the tokens with .word suffix
  separate(token_code, into = c("item", "zone", "type"), sep = "\\.") %>%
  mutate(
    item = as.numeric(item),
    zone = as.numeric(zone),
    log_freq = log(freq_n)
  ) %>%
  select(item, zone, word, log_freq)

print("3. Aligning and merging the data")

## [1] "3. Aligning and merging the data"

final_model_data <- rt_data %>%
  inner_join(freq1_data, by = c("item", "zone", "word")) %>%
  filter(is.finite(log_freq))

print("4. Fitting a mixed-effects regression model...")

## [1] "4. Fitting a mixed-effects regression model..."

model <- lmer(RT ~ log_freq + (1 | WorkerId) + (1 | item),
              data = final_model_data)

summary(model)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: RT ~ log_freq + (1 | WorkerId) + (1 | item)
## Data: final_model_data
##
## REML criterion at convergence: 9362237

```

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##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.3648 -0.4506 -0.1780  0.1656 19.4106
##
## Random effects:
##   Groups   Name                Variance Std.Dev.
##   WorkerId (Intercept)  6648.0    81.54
##   item      (Intercept)   121.1    11.01
##   Residual                    20307.6  142.50
## Number of obs: 733810, groups:  WorkerId, 180; item, 10
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  3.664e+02  7.093e+00  1.049e+02   51.65  <2e-16 ***
## log_freq     -1.864e+00  5.562e-02  7.336e+05  -33.51  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## log_freq -0.146

```